

Q) If inputs $ABCD = 1001, 1011$ never occur, then simplified expression for

$$f(A, B, C, D) = \bar{A}B\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}CD + ABD + ABC$$

Never occur $\Rightarrow$ Don't care $\Rightarrow 9, 11$

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11		1	1	1
10		X	X	0

$$Y = D + ABC$$

(Q) The truth Table is given

A	B	C	Y
0	0	0	1
0	0	1	X ₁
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	X ₂
1	1	0	1
1	1	1	1

To get simplest boolean expression of Y , X_1 & X_2 should be?

A.

BC				
	00	01	11	10
A=0	1	X ₁		
A=1		X ₂	1	1

$$X_2 = 0$$

$$X_1 = 1$$

2. Mappings.

(Q) $f(A, B, C, D) = \sum m(0, 1, 3, 7, 12, 13, 15)$

Find the no. of PI & EPI?

A.

CD				
	00	01	11	10
AB=00	1	1	1	
AB=01				1
AB=11	1	1	1	
AB=10				

2-EPI

6-P.I

$$\sum m(0, 1) \text{ (EPI)}$$

$$\sum m(1, 3)$$

$$\sum m(3, 7)$$

$$\sum m(7, 15)$$

$$\sum m(13, 15)$$

$$\sum m(12, 13) \text{ (EPI)}$$

A.

CD				
	00	01	11	10
AB=00		1		X → 0
AB=01	X	1		1
AB=11	1	1		1
AB=10				

Y (SOP)

$$Y = B\bar{C} + \bar{A}\bar{C}D + B\bar{C}\bar{D} \quad L.C = 7$$

$$F(A, B, C, D) = \prod M(6, 3, 7, 9, 9, 10, 11, 15) \quad \prod d(2, 4)$$

CD				
	00	01	11	10
A+B=00	0		0	X
A+B=01	X		0	
A+B=11			0	
A+B=10	0	0	0	0

(POS)

$$Y = (\bar{A} + B) \cdot (\bar{C} + \bar{D}) \cdot (B + D) \quad L.C = 6$$

(Q) Find the minimum Possible Literal Count.

$$f(A, B, C, D) = \sum m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

(a) 7 (b) 8 (c) 9 (d) 6

Number System				Any Number with base (r)
Decimal	Binary	Octal	Hexa decimal	
[Base 10]	[Base 2]	[Base 8]	[Base 16]	[Base r]
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]	[0, 1]	[0, 1, 2, 3, 4, 5, 6, 7]	[0, 1, 2, ..., 9, A, B, C, D, E, F]	Eg: $[αβγ]_r$
Eg: $[93]_{10} (✓)$	Eg: $[1010]_2 (✓)$	Eg: $[723]_8 (✓)$	Eg: $[A32]_{16} (✓)$	$r > α, β, γ$
$[127]_{10} (✓)$	$[100]_2 (✓)$	$[982]_8 (X)$	$[923]_{16} (✓)$	
$[216]_{10} (X)$	$[216]_2 (X)$	$[777]_8 (✓)$	$[H9D]_{16} (X)$	
$[32]_{10} (X)$				

① Decimal number system :- [Base 10]

[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

$$\begin{aligned} \text{① } [2743]_{10} &= 3 \times 10^0 + 4 \times 10^1 + 7 \times 10^2 + 2 \times 10^3 \\ &= 3 + 40 + 700 + 2000 = 2743_{10} \end{aligned}$$

$$\begin{aligned} \text{② } [43.5]_{10} &= 3 \times 10^0 + 4 \times 10^1 + 5 \times 10^{-1} = 3 + 40 + 0.5 \\ &= 43.5_{10} \end{aligned}$$

$$\text{③ } [973256.5374]_{10}$$

$$\begin{aligned} &= 6 \times 10^0 + 5 \times 10^1 + 2 \times 10^2 + 3 \times 10^3 + 7 \times 10^4 + 9 \times 10^5 + 5 \times 10^6 + \\ &\quad 3 \times 10^{-2} + 7 \times 10^{-3} + 5 \times 10^{-4} \end{aligned}$$

$$= 6 + 50 + 200 + 3000 + 70000 + 900000 + 0.5 + 0.03 +$$

$$0.0007 + 0.0005$$

$$= 973256.5374_{10}$$

② Binary number system :- [Base 2]

[0, 1]

$$\begin{aligned} \text{① } [1101]_2 &= 1 \times 2^0 + 0 \times 2^1 + 1 \times 2^2 + 1 \times 2^3 \\ &= 1 + 0 + 4 + 8 = [13]_{10} \end{aligned}$$

$$\begin{aligned} \text{② } [101.11]_2 &= 1 \times 2^0 + 0 \times 2^1 + 1 \times 2^2 + 1 \times 2^{-1} + 1 \times 2^{-2} \\ &= 1 + 0 + 4 + 0.5 + 0.25 = [5.75]_{10} \end{aligned}$$

$$\begin{aligned} \text{③ } [1101011.101]_2 &= 1 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 0 \times 2^4 + 1 \times 2^5 \\ &\quad + 1 \times 2^6 + 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} \\ &= 1 + 2 + 0 + 8 + 0 + 32 + 64 + 0.5 + 0 + 0.125 \\ &= [107.625]_{10} \end{aligned}$$

③ Octal number system :- [Base 8]

[0, 1, 2, 3, 4, 5, 6, 7]

$$\begin{aligned} \text{① } [6472]_8 &= 2 \times 8^0 + 7 \times 8^1 + 4 \times 8^2 + 6 \times 8^3 \\ &= 2 + 56 + 64 \times 4 + 512 \times 6 \\ &= 58 + 256 + 3072 = [3386]_{10} \end{aligned}$$

$$\textcircled{2} [1011]_8 = 1 \times 8^0 + 1 \times 8^1 + 0 \times 8^2 + 1 \times 8^3$$

$$= 1 + 8 + 512 = [521]_{10} //$$

$$\textcircled{3} [67316.349]_8 = 6 \times 8^0 + 1 \times 8^1 + 3 \times 8^2 + 7 \times 8^3 + 6 \times 8^4 +$$

$$3 \times 8^{-1} + 4 \times 8^{-2} + 9 \times 8^{-3}$$

$$= 6 + 8 + 192 + 3584 + 24576 +$$

$$0.375 + 0.0625 + 0.017578125$$

$$= [28366.45508]_{10} //$$

Hexadecimal numbers System :- [Base 16]

[0, 1, ..., 9, A, B, C, D, E, F]

Ex: $\textcircled{1} [953]_{16} = 3 \times 16^0 + 5 \times 16^1 + 9 \times 16^2$

$$= 3 + 80 + 2304 = [2387]_{10} //$$

$$\textcircled{2} [C549DF]_{16} = 15 \times 16^0 + 13 \times 16^1 + 9 \times 16^2 + 4 \times 16^3$$

$$+ 5 \times 16^4 + 12 \times 16^5$$

$$= 15 + 208 + 2304 + 16384 +$$

$$327680 + 12582912$$

$$= [12929503]_{10} //$$

Conversions :-

Any base to decimal

$$\rightarrow [342.10]_5 = 2 \times 5^0 + 4 \times 5^1 + 3 \times 5^2 + 1 \times 5^{-1} + 0 \times 5^{-2}$$

$$= 2 + 20 + 75 + 0.2 + 0$$

$$= [97.2]_{10} //$$

$$\rightarrow [A27.F.C]_{16} = [15 \times 16^0 + 7 \times 16^1 + 2 \times 16^2 + 10 \times 16^3 +$$

$$12 \times 16^{-1}]$$

$$= [15 + 112 + 512 + 40960 + 0.75]$$

$$= [41599.75]_{10} //$$

$$\rightarrow [8762]_9 = 2 \times 9^0 + 6 \times 9^1 + 7 \times 9^2 + 8 \times 9^3$$

$$= 2 + 54 + 567 + 5832 = [6455]_{10} //$$

Decimal to any Base

$$\rightarrow (78.25)_{10} = (?)_2$$

Int.
Part

2	73	
2	36	1
2	18	0
2	9	0
2	4	1
2	2	0
2	1	0
2	0	0

$(1001001)_2$

$$\begin{array}{l} 0.25 \times 2 = 0.50 \\ 0.50 \times 2 = 1.00 \\ 0.00 \times 2 = 0.00 \\ 0.00 \times 2 = 0.00 \end{array}$$

$(.01)_2$

$$[78.25]_{10} = [1001001.01]_2$$

$$\rightarrow (739.26)_{10} = ()_8$$

$$\begin{array}{r|l} 8 & 739 \\ 8 & 92 \\ 8 & 11 \\ & 1 \end{array}$$

$$\begin{array}{c} 3 \\ 4 \\ 3 \end{array}$$

$$(1343)_8$$

$$.26 \times 8 = 2.08$$

$$0.8 \times 8 = 0.64$$

$$.64 \times 8 = 5.12$$

$$0.12 \times 8 = 0.96$$

$(205)8$

$$\Rightarrow (739.26)_{10} = (1343.205)_{8/4}$$

$$\rightarrow (2932.62)_{10} = (\quad)_{16}$$

16	7932	
16	495	C ↑
16	30	F
	1	E

$$.67 \times 16 = 10.72$$

$$.72 \times 16 = 11.52$$

$$\bullet .52 \times 16 = 8.32$$

$$.32 \times 16 = 5.12$$

$$(1EFC)_{16}$$

$$\cdot (AB85)_{16}$$

$$(7932 \cdot 62)_{16} = (1EFC \cdot AB85)_{16}.$$

(★) Conversion from any base to another Base :-

eg:- $[341.12]_5 = []_6$

→ Convert given one to Base 10.

→ From Base 10 to whichever base you want.

$$\Rightarrow [341.12]_5 = 1 \times 5^0 + 4 \times 5^1 + 3 \times 5^2 + 1 \times 5^3 + 2 \times 5^4$$

$$= 1 + 20 + 75 + 125 + 625$$

$$= [96.28]_{10} \approx (.)_6$$

$$= 1 + 20 + 75 + 0.2 + 0.08$$

$$= [96.28]_{10\%} \approx (-)_{\%}$$

$$\Rightarrow \begin{array}{r|l} 6 & 96 \\ 6 & 16 \\ 6 & 2 \end{array} \quad \begin{array}{l} 0 \\ 4 \end{array} \uparrow$$

$$(240)_6$$

$$.28 \times 6 = 1.68$$

$$\therefore 68 \times 6 = 408$$

$$.08 \times 6 = 0.48$$

$(140)6$

$$\rightarrow [341.12]_5 = [240.140]_{11}$$

(★) special case :-

→ Binary to 2^n Numbers System :- $[n \in \mathbb{N}]$

$$\text{Ex: } (10 \leftrightarrow 11 \leftrightarrow 100 \leftrightarrow 10110 \cdot \leftrightarrow 110110)_2 = ()_8 = ()_{16} = ()_4$$

$$(a) \Rightarrow ()_2 = ()_{2^3}$$

$$= (26626.66)_8. \quad (\text{use 3 grouping})$$

$$(b) \Rightarrow ()_2 = ()_{24} = \underbrace{0010}_{10} \underbrace{110110}_{18} \underbrace{010110}_{22} \underbrace{11011000}_{44}$$

$$= (2096.08)_{16} \quad (\text{use 4 grouping})$$

(Q) $\sqrt{41} = 5$. Find base?

A. $\sqrt{(41)_r} = (5)_r$

$$\Rightarrow \sqrt{1xr^0 + 4xr^1} = 5xr^0$$

$$\Rightarrow \sqrt{1 + 4r} = 5$$

$$\Rightarrow 1 + 4r = 25 \Rightarrow 4r = 24 \Rightarrow \boxed{r=6}$$

(Q) In a Particular Number system, (1) roots of $x^2 - 11x + 22 = 0$ is 3. Find base?

A. root = $(3)_r$

Product of roots = $c/a = (22)_r$

Sum " " = $(11)_r$

Let, other root = $(x)_r$

$$\Rightarrow (3)_r \cdot (x)_r = (22)_r$$

$$\Rightarrow 3(\underline{(x)_r}) = 2xr^0 + 2xr^1 = 2 + r$$

(x say)

$$\Rightarrow 3x = 2 + r \rightarrow \textcircled{1}$$

$$\Rightarrow (3)_r + (x)_r = (11)_r$$

$$\Rightarrow (3)_r + x = 1xr^0 + 1xr^1 = 1 + r$$

$$\Rightarrow 3 + x = 1 + r \rightarrow \textcircled{2}$$

$$\textcircled{2} - \textcircled{1} \Rightarrow 9 + 3x = 3 + 3r$$

$$(-) 2x = 2 + 2r$$

$$9 = 1 + 2r$$

$$3x = 2 + 16$$

$$\Rightarrow x = 18/3 = 6 \text{ (other root)}$$

$$\Rightarrow \boxed{r=8}$$

(Q) $(43)_x = (y3)_8$. Find all possible Comb. of x & y ?

A. $3xx^0 + 4xx^1 = 3 + 8y$

$$\Rightarrow 3 + 4x = 3 + 8y \Rightarrow \boxed{x = 2y}$$

$$\Rightarrow \boxed{y = x/2}$$

$$(43)_x \quad x > 4 \Rightarrow x [5, 6, \dots]$$

$$(y3)_8 \quad y < 8 \Rightarrow y [1, 2, 3, \dots]$$

$$x = 5; y = 5/2 \text{ (x)} \Rightarrow x = 6, y = 3 \checkmark$$

$$x=8 \Rightarrow y=4 (\checkmark)$$

$$x=10 \Rightarrow y=5 (\checkmark)$$

$$x=12 \Rightarrow y=6 (\checkmark)$$

$$x=14 \Rightarrow y=7 (\checkmark)$$

$$x=16 \Rightarrow y=8 (X)$$

(a) Find possible no. of 1's present in binary representation of $(3 \times 512) + (7 \times 64) + 3$.

$$\underline{A.} \quad N = (3 \times 512) + (7 \times 64) + (0 \times 8) + (3 \times 8^0)$$

$$= (3 \times 8^3) + (7 \times 8^2) + (3 \times 8^0)$$

$$= (3703)_8 = (3703)_2$$

$$= (0 \underline{11} \quad \underline{111} \quad 000 \quad 0 \underline{11})_2$$

7 1's.

$$(a) \quad M \times 32768 + J \times 1024 + 3 = (?)_{82}$$

$$\underline{A.} \Rightarrow M \times (32)^3 + J \times (32)^2 + 0 \times (32)^1 + 3 \times (32)^0$$

$$\Rightarrow (MJ03)_{32}$$

(a) $\frac{41}{3} = 13$ in some Number System. Find Base?

$$\underline{A.} \quad \frac{(41)_r}{(3)_r} = (13)_r \Rightarrow \frac{4r+1}{3} = r+3$$

$$\Rightarrow 4r+1 = 3r+9$$

$$\Rightarrow \boxed{r=8}$$

M22)

$$(a) \quad \frac{55}{5} = 11. \text{ Possible base is}$$

$$(\checkmark) 10 \quad (b) 3 \quad (c) 5 \quad (\checkmark) 6$$

$$\underline{A.} \quad \frac{(55)_r}{(5)_r} = (11)_r \Rightarrow \boxed{r > 5} \Rightarrow \begin{array}{|l} \text{For a } r \text{ base} \\ \text{even, digits are} \\ \text{0 to } (r-1) \end{array}$$

$$\Rightarrow \frac{5r+5}{5} = r+1 \Rightarrow \boxed{5r+5 = 5r+5}$$

$$[r > 5]$$

END OF

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